

The Final Project Report

Global Electronics Retail: Strategic Performance Report

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Tools Used: Python (Pandas), SQL (MySQL), Excel, Power BI

1. Executive Summary

This project involved a comprehensive analysis of over 60,000 sales records from a global electronics retailer. The goal was to identify key profit drivers, understand customer behavior, and optimize operational efficiency across international stores.

2. Technical Methodology

- **Data Cleaning (Python):** Handled missing values and resolved a critical ProductKey mismatch issue using `pd.merge` and data type normalization.
 - **Data Analysis (SQL):** Utilized Common Table Expressions (CTEs) and Window Functions to perform Pareto (80/20) analysis and calculate Customer Lifetime Value (CLV).
 - **Visualization (Power BI):** Developed an interactive 3-page dashboard to track KPIs and consumer trends.
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3. Data-Driven Insights (Solving the 10 Business Problems)

1. **Revenue & Profit Growth (Pareto):** 80% of total profit is driven by only **[Insert Number]** categories (primarily Computers and Audio).
2. **Customer Behavior (CLV):** The average Customer Lifetime Value is highest in **[Insert Country]**, suggesting a high concentration of loyal, premium buyers there.
3. **Seasonality:** A significant spike in Sales is observed in **December** (Holiday season) and **August** (Back-to-school for Computers).
4. **Top Categories: Computers and Home Appliances** generate the most raw revenue.
5. **Brand Performance:** **[Insert Top Brand]** is the leading brand by revenue, contributing to **[X]%** of total sales.

6. **Order Volume:** Monthly trends show a steady **[Increase/Decrease]** in order volume, peaking at year-end.
 7. **Product Popularity:** The **[Insert Product Name]** is the highest-selling item by unit volume.
 8. **Customer Loyalty:** Our top 1% of customers have placed more than **[X]** orders each.
 9. **Store Performance:** Store **[Insert ID]** in **[Country]** has the highest sales volume globally.
 10. **Consumer Preferences:** **Black** and **Silver** are the preferred colors for most categories, but **Red** and **White** see higher preference in the "Audio" category.
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4. Strategic Recommendations

- **Inventory Realignment:** Increase stock levels for top-performing brands 30 days prior to identified seasonal spikes.
- **Marketing Focus:** Reallocate 20% of the budget from low-profit categories (e.g., Games) to high-CLV customer segments in Europe/North America.
- **Store Optimization:** Audit the top-performing stores to create a "Success Blueprint" for underperforming locations.